forces acting on a particle. Initially we set elements to $\bar{O}$, the null vector.

#6 Iterate over each particle.

#7 For a particle, find its neighbors, i.e. the other particles whose effects will be more relevant on current one. In this example, we use a spherical range search within a certain radius, f.i. a threshold at which interaction is not relevant anymore. This value depends on the problem, obviously. Alternatively, we could also decide to only compute the interaction with the `m` closest points, for some `m`.

#8 For each of the selected “neighbors”, compute the force resulting from the interaction of the two particles.

#9 In this configuration, we compute the force between particles `A` and `B` twice: once when `particle==A` and `neighbor==B`, and once when `particle==A` and `neighbor==B`. With a small change we can keep track of the pairs updated and make sure to compute the interaction only once.

#10 Once all forces are computed, cycle through all particles again and update their position and velocity. We should also take into account interaction with the box boundaries, for instance inverting the velocity if the particle hits the walls, assuming an elastic collision (or more complex interaction in case of inelastic collision).

Moreover, it doesn’t even always make sense: electrical force and gravity have a limited range of action, and so outside of a certain radius the magnitude of the interaction between two particles would be negligible.